ELEG3124
Homework Assignment
Lab1

1- Write the set of commands that are required to create the following matrices: (The answers are not unique)

$$\begin{bmatrix}1 & 1 & 1 & 1\\ 1 & 0 & 0 & 1\\ 1 & 0 & 0 & 1\\ 1 & 1 & 1 & 1\end{bmatrix} \quad \begin{bmatrix}0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 1 & 1\\ 0 & 0 & 1 & 1\end{bmatrix} \quad \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

2- For the following matrices first calculate the normal and element-wise matrix multiplication in MATLAB and then do it by hand and show why the normal and element-wise multiplication of matrices give different results:

$$A = \begin{bmatrix}+1 & +2\\ -1 & +5\end{bmatrix} \qquad B = \begin{bmatrix}+2 & +1\\ +6 & +3\end{bmatrix}$$

3- For each of the following continuous functions plot the function using both normal plot (plot) and symbolic plot (ezplot) commands : (Assume x to be between 0 to 20 with increments of 0.1)

a. $y = \sin(x) + \cos(x)$
b. $y = e^{x} + x + 1$
c. $y = \ln(2 + x + x^{2})$
d. $y = \sec(x)^{2} + \cot(x)$

4- For the following discrete functions calculate the first 20 elements (i.e. n = 0…20) and plot them by using the discrete signal plotting command (stem):

a. $y = 10n^{2}\sin(n)$
b. $y = \sin(0.5n)/n$

5- Write an m-file (both as a script and as a function) which gets an input N from the user and shows all the prime numbers less than or equal to N.

Note: Do **NOT** use the **primes** function which is already defined in MATLAB.
Note: Name your m-files as **show_primes.m** and **primes_func.m** and assume **N > 1**.

Good Luck

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